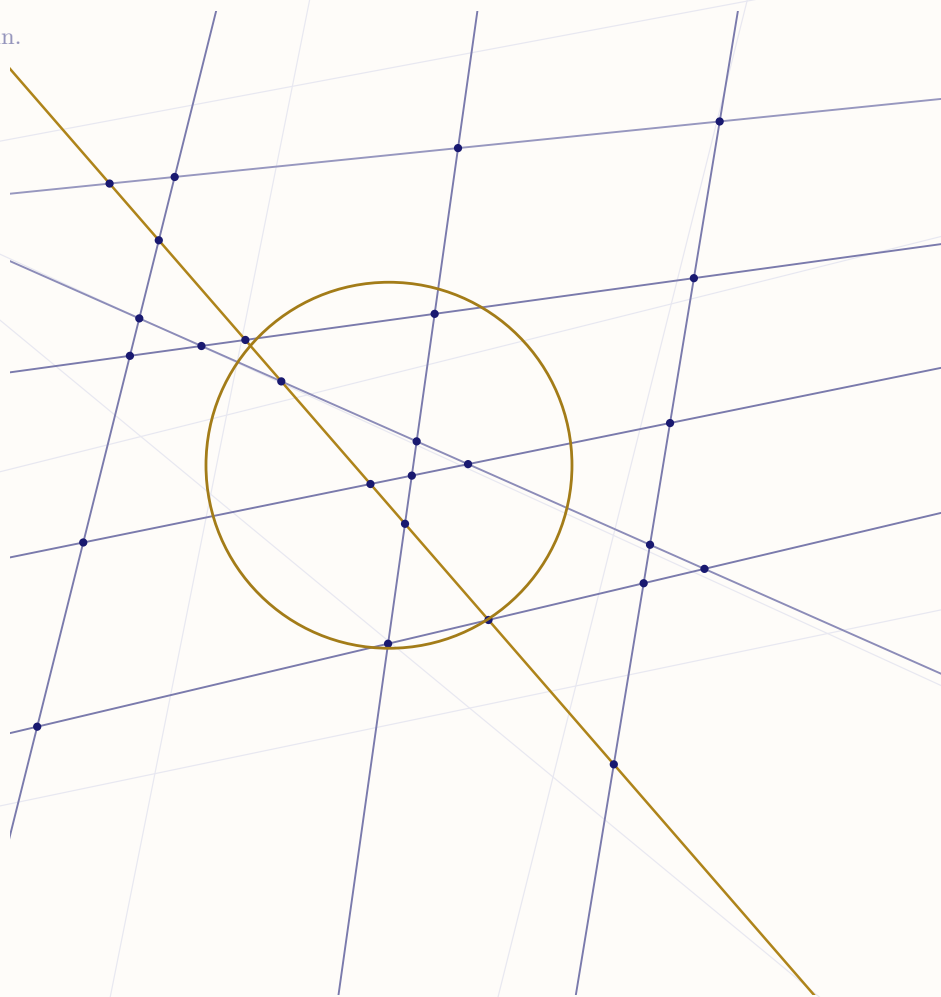


Matroid Theory

after Federico Ardila

independence · duality · representation

Rewritten from the handwritten 2007 course.
Diagrams restored; incomplete arguments filled in.



a rank-3 line arrangement

$\mathcal{I} \cdot \mathcal{B} \cdot \mathcal{C} \cdot r \cdot \text{cl} \cdot L(M)$

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About these notes

These notes follow Federico Ardila's handwritten 2007 course ([lectures1-25.pdf](#)). The material is rewritten in full and arranged by topic rather than by lecture date. Diagrams from the board notes are redrawn in TikZ. Proofs that were only a sketch on the board are written out.

Editorial marks are coloured words in the text itself: **blue** for a correction of a mistake or an imprecise statement in the scans; **green** for an argument that was omitted, left as an exercise, or written too sketchily on the board; **purple** for a remark that goes beyond the 2007 notes (standard facts, later theorems, or a short development of a theme the lectures only mention). Purple passages include uniform matroids, Hall's theorem, lattice path matroids, direct sums, the Tutte polynomial, Cauchy–Binet and matroid intersection, strong circuit elimination, Whitney planarity, the cryptomorphism dictionary, the scum theorem, binary and regular matroids, Pappus and circuit-hyperplane relaxation, and the comparison of linear with algebraic dependence.

A single six-element matroid runs through the notes: it is realised at once by a vector configuration, a graph, and a bipartite matching.

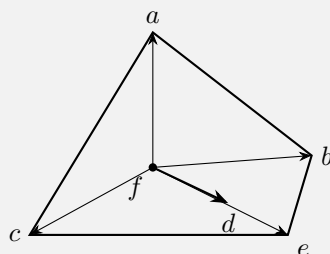
1 Three pictures of independence

The same combinatorial pattern appears in linear algebra, graph theory, and matching theory. Throughout these notes we keep a single six-element running example that realises all three pictures at once.

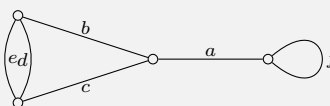
Example 1.1 (Linear algebra). Let $E = \{a, b, c, d, e, f\}$ be the list of vectors in $\mathbb{R}^3$

$$a = (1, 0, 0), \quad b = (0, 1, 0), \quad c = (0, 0, 1), \quad d = (0, \tfrac{1}{2}, \tfrac{1}{2}), \quad e = (0, 1, 1), \quad f = (0, 0, 0).$$

Goal: choose a linearly independent subset of E .



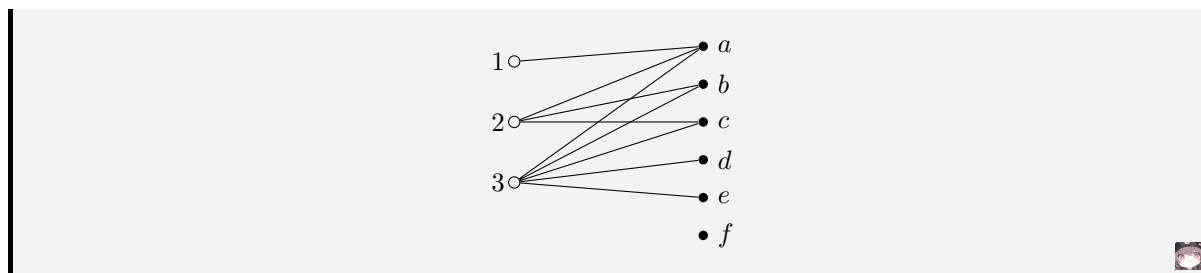
Example 1.2 (Graph theory). Let G be the graph on ground set $E = \{a, b, c, d, e, f\}$ drawn below (f is a loop, d and e are parallel, and a is a pendant edge). Goal: choose a set of edges containing no cycle.



Example 1.3 (Matching theory). Let G be the bipartite graph with left vertices $\{1, 2, 3\}$ and right vertices $\{a, b, c, d, e, f\}$, with neighbourhoods

$$N(1) = \{a\}, \quad N(2) = \{a, b, c\}, \quad N(3) = \{a, b, c, d, e\}.$$

(The isolated vertex f is unmatched; the original notes add a joke which we omit.) Goal: choose a set of right-hand vertices that can be matched to distinct left-hand vertices.



1.1 What is special about the elements?

Comparing the three models on the same ground set:

- Q1. One element is unlike the others: f is *useless*. It is the zero vector, a loop, and an unmatched vertex. One can never choose it.
- Q2. Another element is unlike the others: a is *essential*. It is the only vector with a nonzero first coordinate, a bridge (isthmus) in the graph, and the unique neighbour of vertex 1. One should always choose it when building a maximal independent set.
- Q3. Two further elements are unlike the others: d and e are *twins* (parallel). In the vector picture $e = 2d$; in the graph they are parallel edges; in the matching they are interchangeable. One may choose at most one of them.

Matroid theory is the axiomatic study of this notion of independence. The three examples will turn out to be *the same matroid*.

Whitney introduced matroids in 1935 to isolate the combinatorial content of linear dependence, and noticed at once that graphs supply a second model. The later matching and algebraic models sit on the same axioms. The word “matroid” is meant to sound like “matrix”: a matroid is what remains of a matrix after one forgets the field and remembers only which columns are independent.

2 Independent sets

2.1 The running example

Example 2.1 (Independent sets of vectors). For the configuration of Example 1.1, the independent sets are

$$\begin{aligned} &\emptyset, \\ &\{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \\ &\{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}, \{b, c\}, \{b, d\}, \{b, e\}, \{c, d\}, \{c, e\}, \\ &\{a, b, c\}, \{a, b, d\}, \{a, b, e\}, \{a, c, d\}, \{a, c, e\}. \end{aligned}$$

In particular f appears in none of them, and $\{d, e\}$ is missing.

Exercise 2.2. Check that the acyclic edge-sets of Example 1.2 and the matchable subsets of Example 1.3 are *exactly* the same collection.

Write $M(G)$ for the cycle matroid of the graph in Example 1.2. The unique loop is f . The unique 2-circuit is $\{d, e\}$. The 3-circuits are $\{b, c, d\}$ and $\{b, c, e\}$. An edge-set is acyclic if and only if it contains none of these, which is the list above.

In the matching picture a set A of right-hand vertices is matchable if and only if it satisfies Hall's condition with respect to $\{1, 2, 3\}$. Direct inspection again yields the same list: f is never matchable, $\{d, e\}$ is not (both can only be reached from vertex 3), and the maximal matchable sets are $\{a, b, c\}$, $\{a, b, d\}$, $\{a, b, e\}$, $\{a, c, d\}$, $\{a, c, e\}$.

Matroid theory studies “independence”; in particular, it studies the common structure of Examples 1.1–1.3.

2.2 The independent-set axioms

Definition 2.3 (Matroid). A *matroid* is a pair $M = (E, \mathcal{I})$ where E is a finite *ground set* and $\mathcal{I} \subseteq 2^E$ is a collection of *independent sets* satisfying:

- (I1) $\emptyset \in \mathcal{I}$.
- (I2) If $I \subseteq J$ and $J \in \mathcal{I}$, then $I \in \mathcal{I}$.
- (I3) If $I, J \in \mathcal{I}$ and $|I| < |J|$, then there exists $j \in J \setminus I$ such that $I \cup \{j\} \in \mathcal{I}$.



The scans write (I2) with a proper inclusion $I \subset J$. The correct axiom is $I \subseteq J$ (which is automatic if $I = J$). We use $\subseteq$ throughout.

Axiom (I3) is the *augmentation* (or independent-set exchange) axiom.

2.3 The three models are matroids

Proposition 2.4 (Linear matroids). Let E be a finite list of vectors in a vector space V , and let $\mathcal{I}$ be the collection of linearly independent sublists. Then $(E, \mathcal{I})$ is a matroid.



Proof. (I1) and (I2) are immediate. For (I3), let $I, J \in \mathcal{I}$ with $|I| < |J|$, and suppose for contradiction that $I \cup \{j\} \notin \mathcal{I}$ for every $j \in J \setminus I$. Write $I = \{v_1, \dots, v_k\}$. For each such j the set $I \cup \{j\}$ is dependent, so there is a linear relation

$$c_1 v_1 + \dots + c_k v_k + c_j j = 0$$

with not all coefficients zero. The coefficient c of j cannot vanish (else I would be dependent), hence $j \in \text{span}(I)$. Thus $J \subseteq \text{span}(I)$, so $\text{span}(J) \subseteq \text{span}(I)$ and

$$|J| = \dim \text{span}(J) \leq \dim \text{span}(I) = |I|,$$

where the two equalities use that I and J are independent. This contradicts $|I| < |J|$. □

Proposition 2.5 (Graphic matroids). Let $G = (V, E)$ be a graph (loops and multiple edges allowed), and let $\mathcal{I}$ be the collection of acyclic subsets of E . Then $(E, \mathcal{I})$ is a matroid, written $M(G)$ and called the *cycle matroid* (or *graphic matroid*) of G .



Proof. (I1) and (I2) are immediate. The key observation is:

Observation 2.6. If $I \subseteq E$ is acyclic, then the spanning subgraph (V, I) has exactly $|V| - |I|$ connected components.



Indeed, a forest on n vertices with c components has $n - c$ edges.

Now let $I, J \in \mathcal{I}$ with $|I| < |J|$, and suppose $I \cup \{j\} \notin \mathcal{I}$ for all $j \in J \setminus I$. Then each such j forms a cycle with I , so adding j does not connect two components of (V, I) . Hence $(V, I \cup J)$ still has $|V| - |I|$ components, and therefore so does (V, J) . But J is acyclic, so Observation 2.6 gives

$$|V| - |J| = \text{number of components of } (V, J) \geq |V| - |I|,$$

hence $|J| \leq |I|$, a contradiction. □

Proposition 2.7 (Transversal matroids). Let G be a bipartite graph with bipartition (D, E) , and let $\mathcal{I}$ be the collection of subsets of E that can be matched into D . Then $(E, \mathcal{I})$ is a matroid.



Proof. The notes leave this as Homework 1. We record a complete argument.

(I1) and (I2) are immediate (the empty set is matched by the empty matching, and a subset of a matched set is matched). For (I3), let $I, J \subseteq E$ be matchable, with $|I| < |J|$, and fix matchings M_I of I and M_J of J into D . Consider the symmetric difference $M_I \triangle M_J$, a disjoint union of even cycles and alternating paths. Because $|J| > |I|$, some path P in $M_I \triangle M_J$ is M_I -augmenting: if $j \in J \setminus I$ is the E -endpoint of P , then $M_I \triangle P$ is a matching of $I \cup \{j\}$. □

2.4 Isomorphism, and names for the three classes

Definition 2.8 (Isomorphism). Two matroids $(E, \mathcal{I})$ and $(E', \mathcal{I}')$ are *isomorphic* if there is a bijection $E \rightarrow E'$ sending $\mathcal{I}$ onto $\mathcal{I}'$. We often simply say they are “the same matroid”.



The three examples of Examples 1.1 to 1.3 are isomorphic; we treat them as one running example.

Definition 2.9. A matroid is *linear* (or representable over a field $\mathbb{F}$) if it is isomorphic to the matroid of a vector configuration in an $\mathbb{F}$ -vector space. It is *graphic* if it is isomorphic to $M(G)$ for some graph G . It is *transversal* if it arises from a bipartite matching problem as in Proposition 2.7.



A fourth basic class, used constantly below: the *uniform matroid* $U_{r,n}$ on an n -set has as independent sets all subsets of size at most r . Thus $U_{n,n}$ is free (every set independent), $U_{0,n}$ consists of n loops, and $U_{1,n}$ is n parallel non-loop elements. A generic vector configuration of n vectors in $\mathbb{F}^r$ realises $U_{r,n}$ whenever $|\mathbb{F}|$ is large enough.

The special elements of Lecture 1 have official names. An element e is a *loop* if $\{e\}$ is dependent, a *coloop* (isthmus) if e lies in every basis, and two non-loop elements are *parallel* if they form a 2-circuit. In the running example f is a loop, a is a coloop, and $\{d, e\}$ is a parallel pair.

In a graphic matroid the rank of an edge-set A is $r(A) = |V| - \kappa(A)$, where $\kappa(A)$ is the number of connected components of (V, A) , counting isolated vertices. This is the same counting that made (I3) work in Proposition 2.5.

Hall's marriage theorem supplies a direct test for independence in a transversal matroid: $A \subseteq E$ is matchable if and only if $|N(S)| \geq |S|$ for every $S \subseteq A$, where $N(S)$ is the set of left-hand neighbours of S . The running example fails Hall on $\{d, e\}$ (both meet only vertex 3) and on any set containing f .

The philosophical list in the bases chapter mentions Dyck paths as a fourth model of independence. A *lattice path matroid* (Bonin–de Mier–Noy) has ground set the north-steps of lattice paths from $(0, 0)$ to (m, r) lying between two bounding paths; a set of north-steps is independent if it occurs in some such path. The *Catalan matroid* is the special case whose bounds are the axes and the diagonal: its bases correspond to Dyck paths of semilength n (rank n on $2n$ elements). Lattice path matroids are transversal, hence representable over every large enough field.

3 Bases

Definition 3.1 (Basis). A *basis* of a matroid is a maximal independent set: an independent set not properly contained in a larger independent set.



In the running example

$$\mathcal{B} = \{abc, abd, abe, acd, ace\}.$$

This is a shorter description of M , because the independent sets are precisely the subsets of bases.

Proposition 3.2. All bases of a matroid have the same cardinality.



Proof. Suppose B_1, B_2 are bases with $|B_1| < |B_2|$. By (I3) there is $b \in B_2 \setminus B_1$ with $B_1 \cup \{b\}$ independent, contradicting maximality of B_1 . $\square$

The common cardinality is the *rank* $r(M)$ of M .

Corollary 3.3. Every independent set is contained in a basis.



Proof. Let I be independent. Among independent sets containing I , take one of maximal cardinality; it is a basis by Proposition 3.2, else (I3) would enlarge it. $\square$

Example 3.4. In the matching picture one cannot match all six right-hand vertices. But every matchable set extends to a matchable set of size $r(M) = 3$. For instance $\{a, d\}$ extends (e.g. to $\{a, b, d\}$).



Example 3.5 (Linear). In a linear matroid a basis is a basis of the span, and equicardinality is dimension. 

Example 3.6 (Graphic). If G is connected, the bases of $M(G)$ are the spanning trees of G (connected, acyclic, spanning subgraphs). If G is disconnected, they are the spanning forests (a spanning tree of each component). That these all have the same number of edges is the matroidal content of “ $n - c$ edges in a forest with c components”. 

Example 3.7 (Transversal). Let S be a finite set with distinguished subsets $S_1, \dots, S_n$. A *transversal* (or system of distinct representatives) is a subset $T = \{a_1, \dots, a_n\} \subseteq S$ with $a_i \in S_i$ for each i . The bases of the corresponding transversal matroid are precisely the transversals. In the running example $S = \{a, b, c, d, e, f\}$ with

$$S_1 = \{a\}, \quad S_2 = \{a, b, c\}, \quad S_3 = \{a, b, c, d, e\},$$

and the transversals are exactly the five bases listed above. 

3.1 Basis axioms

Proposition 3.8. Let E be finite and $\mathcal{B} \subseteq 2^E$. Then $\mathcal{B}$ is the collection of bases of a matroid on E if and only if

(B1) $\mathcal{B} \neq \emptyset$;

(B2) if $A, B \in \mathcal{B}$ and $a \in A \setminus B$, then there exists $b \in B \setminus A$ such that $(A \setminus \{a\}) \cup \{b\} \in \mathcal{B}$. 

Axiom (B2) is *basis exchange*.

Proof that bases of a matroid satisfy (B1)–(B2). Independent sets exist ((I1)), so maximal ones exist: (B1). For (B2), take bases A, B and $a \in A \setminus B$. Then $A \setminus \{a\}$ is independent of size $r(M) - 1$, while B is independent of size $r(M)$. By (I3) there is $b \in B \setminus (A \setminus \{a\})$ with $(A \setminus \{a\}) \cup \{b\}$ independent, hence a basis. This b cannot lie in A (else we would recover A , but $a \notin B$ so $b \neq a$ forces $b \in B \setminus A$). $\square$

Proof that (B1)–(B2) produce a matroid. Let $\mathcal{B}$ satisfy (B1)–(B2). Write

$$\mathcal{I} = \{I \subseteq E : I \subseteq B \text{ for some } B \in \mathcal{B}\}.$$

Step 1. All members of $\mathcal{B}$ have the same size.

Take $B \in \mathcal{B}$ of minimal cardinality. We claim every $A \in \mathcal{B}$ has $|A| = |B|$. Induct on $|A \setminus B|$. If $|A \setminus B| = 0$ then $A \subseteq B$; minimality forces $A = B$. If $|A \setminus B| = k \geq 1$, pick $a \in A \setminus B$. By (B2) there is $b \in B \setminus A$ with $A' = (A \setminus \{a\}) \cup \{b\} \in \mathcal{B}$. Then

$$|A' \setminus B| = |A \setminus B| - 1,$$

so the inductive hypothesis gives $|A'| = |B|$. But $|A'| = |A|$, hence $|A| = |B|$.

Write r for this common size.

Step 2. $(E, \mathcal{I})$ is a matroid.

(I1): $\emptyset \subseteq B$ for any $B \in \mathcal{B}$. (I2) is immediate from the definition of $\mathcal{I}$.

For (I3), let $I, J \in \mathcal{I}$ with $|I| < |J|$. We induct on $r - |J|$.

Initial case $r - |J| = 0$: then J is a basis. Let A be a basis containing I , and write

$$X = A \cap J \setminus I, \quad Y = A \setminus J.$$

The scans label two subsets B, C of A without defining them consistently; the three cases below are the intended partition.

- If $X \neq \emptyset$, pick $a \in X$. Then $a \in J \setminus I$ and $I \cup \{a\} \subseteq A$, so $I \cup \{a\} \in \mathcal{I}$.
- If $Y \neq \emptyset$, pick $a \in Y \subseteq A \setminus J$. By (B2) there is $j \in J \setminus A$ with $A' = (A \setminus \{a\}) \cup \{j\} \in \mathcal{B}$. Then $I \cup \{j\} \subseteq A'$ (since $a \notin I$: otherwise $a \in I \subseteq J$, contradicting $a \in Y$) and $j \in J \setminus I$.

- If $X = Y = \emptyset$, then $A \subseteq J$. But $|A| = |J| = r$, so $A = J$ and therefore $I \subseteq J$ with $|I| < |J| = r = |A|$, contradicting that A was a basis containing I unless I can still be augmented—in fact $A = J$ and $I \subsetneq A$, so $A \setminus I \neq \emptyset$ forces $X \neq \emptyset$. This case cannot occur.

Inductive step. Suppose the claim holds whenever $r - |J'| = k - 1$, and now $r - |J| = k \geq 1$. Let $I \subseteq A \in \mathcal{B}$ and $J \subseteq B \in \mathcal{B}$. Applying the initial-case argument to the pair (I, B) (here B is a basis) produces $b \in B \setminus I$ with $I \cup \{b\} \in \mathcal{I}$.

If $b \in J$ we are done. If $b \notin J$, then $J \cup \{b\} \subseteq B$, so $J \cup \{b\} \in \mathcal{I}$. Now $I \cup \{b\}$ and $J \cup \{b\}$ are independent with

$$|I \cup \{b\}| < |J \cup \{b\}|, \quad r - |J \cup \{b\}| = k - 1.$$

The inductive hypothesis supplies $c \in (J \cup \{b\}) \setminus (I \cup \{b\})$ with $I \cup \{b, c\} \in \mathcal{I}$, hence $c \in J \setminus I$ and $I \cup \{c\} \in \mathcal{I}$. $\square$

Definition 3.9 (Matroid via bases). *Equivalently, a matroid is a pair $(E, \mathcal{B})$ with $\mathcal{B}$ satisfying (B1)–(B2). One passes between the two presentations by*

$$\mathcal{B} = \{\text{maximal members of } \mathcal{I}\}, \quad \mathcal{I} = \{\text{subsets of members of } \mathcal{B}\}.$$



3.2 A philosophical aside

Two axiom systems describe the same objects: (I1)–(I3) for independence, and (B1)–(B2) for bases. Proving they are equivalent takes work. The deeper difficulty is to *find* good axiom systems.

(I1)–(I3) is a good system for independence: it applies at once to linear algebra, graphs, matchings (and, later, Dyck paths, ...), and it is precisely the class of simplicial complexes on which greedy works. (B1)–(B2) is a good system for bases: it is weak enough that independence implies it, and strong enough to recover independence.

Direct sums are the disjoint-union operation on matroids. If M and N have disjoint ground sets, the *direct sum* $M \oplus N$ has ground set $E(M) \sqcup E(N)$ and independent sets $I \sqcup J$ with I independent in M and J independent in N . Rank, circuits, and flats all act componentwise: $r_{M \oplus N}(X) = r_M(X \cap E(M)) + r_N(X \cap E(N))$. A matroid is *connected* if it is not a direct sum of two nonempty matroids; equivalently, every pair of elements lies in a common circuit. The running example is connected (after deleting the loop f it remains connected). Direct sum is the operation that later produces the non-representable matroid $F_7 \oplus F_7^*$.

4 The greedy algorithm

4.1 Aside: the matrix-tree theorem

How many spanning trees does a graph have? Form the $V \times V$ *Laplacian* L by

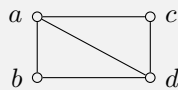
$$L_{vv} = \deg(v), \quad L_{uv} = -(\text{number of edges from } u \text{ to } v) \quad (u \neq v).$$

The eigenvalues of L are $0 = \lambda_0, \lambda_1, \dots, \lambda_{n-1}$.

Theorem 4.1 (Kirchhoff). *The number $\tau(G)$ of spanning trees of G equals $\frac{1}{n} \lambda_1 \cdots \lambda_{n-1}$, and also equals any cofactor of L .*



Example 4.2. For the graph K_4 minus one edge (vertices a, b, c, d , missing bc):



$$L = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}.$$



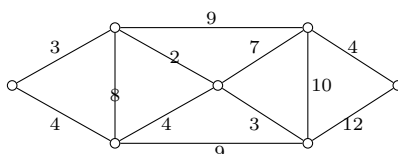
A cofactor computation gives $\tau(G) = 8$. The scan's 3×3 arithmetic is hard to parse. Deleting the first row and column leaves $\det \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 3 \end{pmatrix} = 2(6 - 1) - (-1)(0 + 2) = 10 - 2 = 8$.



 The notes suggest, as a final project, understanding this and its generalization to linear matroids (the Kirchhoff polynomial / basis generating function of a representable matroid).

4.2 Minimum-weight bases

Given a graph $G = (V, E)$ and a weight $w: E \rightarrow \mathbb{R}_{\geq 0}$, find a minimum-weight spanning tree. A naïve-looking strategy works.



Definition 4.3 (Greedy algorithm). Start with $I = \emptyset$. Repeatedly add to I a cheapest element e for which $I \cup \{e\}$ remains independent, until I is a basis.



For graphs this is Kruskal's algorithm (1956). For general matroids the same procedure is due to Rado (1957) and Edmonds (1971); the notes attribute the matroid statement to Kruskal as well.

Proposition 4.4. Let $M = (E, \mathcal{I})$ be a matroid and $w: E \rightarrow \mathbb{R}$ any weight function. The greedy algorithm returns a basis of minimum weight.



Proof. Let $I = \{i_1, \dots, i_r\}$ be the greedy basis, in the order chosen, so $w(i_1) \leq \dots \leq w(i_r)$. Let $J = \{j_1, \dots, j_r\}$ be a minimum-weight basis, ordered $w(j_1) \leq \dots \leq w(j_r)$. Suppose $w(I) > w(J)$, and let k be the least index with $w(i_k) > w(j_k)$.

The sets $\{i_1, \dots, i_{k-1}\}$ and $\{j_1, \dots, j_k\}$ are independent with unequal sizes, so (I3) supplies some j_s with $1 \leq s \leq k$ such that $\{i_1, \dots, i_{k-1}, j_s\}$ is independent. Then

$$w(j_s) \leq w(j_k) < w(i_k),$$

so when the algorithm chose i_k it could already have chosen the cheaper legal element j_s , a contradiction. $\square$

If several elements have equal weight, any cheapest legal choice works: the proof only needs $w(j_s) \leq w(j_k) \leq w(i_k)$, and a strict inequality $w(I) > w(J)$ forces a strict inequality at the first differing position after ties are broken consistently.

4.3 A cryptomorphism: greedy characterises matroids

A *simplicial complex* on E is a pair $(E, \mathcal{I})$ satisfying (I2) (and usually (I1)). Matroids are exactly the simplicial complexes on which greedy optimisation works.

Proposition 4.5. A pair $M = (E, \mathcal{I})$ is a matroid if and only if it satisfies (I1), (I2), and (I3') for every $w: E \rightarrow \mathbb{R}$, the greedy algorithm returns a maximum-cardinality independent set of minimum weight.



Proof. The forward direction is Proposition 4.4. Conversely, assume (I1), (I2), (I3'), and suppose $I, J \in \mathcal{I}$ with $|I| < |J|$ violate (I3): $I \cup \{e\} \notin \mathcal{I}$ for all $e \in J \setminus I$. Let r be the maximum cardinality of an independent

set (greedy produces such a set). Define

$$w(e) = \begin{cases} 1 & e \in I, \\ 2 & e \in J \setminus I, \\ N & e \notin I \cup J, \end{cases} \quad \text{with } N > 2|E|.$$

Greedy first takes all of I (weight 1), then cannot take any element of J , then fills the remaining $r - |I|$ places from $E \setminus (I \cup J)$. The resulting weight is $|I| + N(r - |I|)$.

On the other hand J is independent, so (I2) and maximality of r together with (I3') applied to the zero weighting, which forces all maximum independent sets to have size r yield a maximum independent set $B \supseteq J$ of size r , with weight at most $2|J| + N(r - |J|)$. Then

$$\begin{aligned} 2|J| + N(r - |J|) &< N + N(r - |J|) = N(r - |J| + 1) \\ &\leq N(r - |I|) \leq |I| + N(r - |I|), \end{aligned}$$

since $|I| \leq |J| - 1$. Thus greedy did not find a minimum-weight maximum independent set, contradicting (I3'). $\square$

The number $\tau(G)$ of spanning trees is the evaluation $T_{M(G)}(1, 1)$ of the *Tutte polynomial*

$$T_M(x, y) = \sum_{A \subseteq E} (x - 1)^{r(E) - r(A)} (y - 1)^{|A| - r(A)}.$$

It is the universal deletion–contraction invariant: if e is neither a loop nor a coloop then $T_M = T_{M \setminus e} + T_{M/e}$, while a coloop contributes a factor of x and a loop a factor of y . Specialising recovers the chromatic polynomial of a graph, the reliability polynomial, and (for a linear matroid) the weight enumerator of the associated linear code. The Kirchhoff polynomial mentioned in the notes is the homogeneous generating function $\sum_{B \in \mathcal{B}} \prod_{e \in B} x_e$, of which $\tau(G)$ is the all-ones evaluation.

The final project on the board—extending Kirchhoff from graphs to linear matroids—is Cauchy–Binet. If a rank- r matroid is represented by an $r \times n$ matrix A over $\mathbb{R}$, then

$$\det(AA^T) = \sum_B (\det A_B)^2,$$

the sum running over r -subsets of columns; the nonzero terms are precisely the bases, and $\det A_B \neq 0$ is the Plücker coordinate of B . For the (reduced) incidence matrix of a connected graph this identity is the matrix-tree theorem.

The same augmentation that makes greedy work yields Edmonds' *matroid intersection* theorem: for two matroids M, N on the same ground set,

$$\max\{|I| : I \in \mathcal{I}(M) \cap \mathcal{I}(N)\} = \min_{X \subseteq E} (r_M(X) + r_N(E \setminus X)).$$

Common independent sets of size k in a pair of transversal matroids are matchings in a bipartite graph, so this contains König and Menger.

5 Circuits

We have described matroids by bases and by independent sets. What about dependent sets?

Relabel the running example as $E = \{1, 2, 3, 4, 5, 6\}$ via $a \mapsto 1, \dots, f \mapsto 6$, so

$$\mathcal{B} = \{123, 124, 125, 134, 135\}.$$

The collection $\mathcal{D}$ of dependent sets is long ($\{6\}$, every set containing 6, $\{4, 5\}$, $\{2, 3, 4\}$, $\{2, 3, 5\}$, $\{2, 4, 5\}$, $\{3, 4, 5\}$, ...). It is enough to list the *minimal* dependent sets:

$$\mathcal{C} = \{6, 45, 234, 235\}.$$

Definition 5.1 (Circuit). A *circuit* of a matroid is a minimal dependent set.



Circuits determine the matroid: a set is dependent if and only if it contains a circuit, and independent if and only if it contains none.

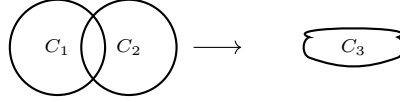
5.1 Circuit axioms

(C1) $\emptyset$ is not a circuit.

(C2) No circuit properly contains another circuit.

(C3) (elimination) If C_1, C_2 are distinct circuits and $x \in C_1 \cap C_2$, then there is a circuit $C_3 \subseteq (C_1 \cup C_2) \setminus \{x\}$.

In graphs, (C3) is the usual cycle-exchange picture: two cycles sharing an edge yield a third cycle in the symmetric difference.



Theorem 5.2. Let E be finite and $\mathcal{C} \subseteq 2^E$. Then $\mathcal{C}$ is the collection of circuits of a matroid on E if and only if it satisfies (C1)–(C3).



Proof of “ $\Rightarrow$ ”. Let $\mathcal{C}$ be the circuits of a matroid. (C1) holds because $\emptyset$ is independent. (C2) is minimality.

For (C3), suppose C_1, C_2 are circuits, $x \in C_1 \cap C_2$, and (for a contradiction) $(C_1 \cup C_2) \setminus \{x\}$ is independent. Since $C_1 \not\subseteq C_2$ there is $y \in C_1 \setminus C_2$. Then $C_1 \setminus \{y\}$ is independent, as is $(C_1 \cup C_2) \setminus \{x\}$. Repeated augmentation of $C_1 \setminus \{y\}$ by elements of C_2 produces, after finitely many steps, the conclusion that $C_1 \cup C_2 \setminus \{y\}$ is independent. But $C_2 \subseteq C_1 \cup C_2 \setminus \{y\}$, so C_2 is independent, a contradiction. $\square$

Define $\mathcal{I} = \{I \subseteq E : \text{no member of } \mathcal{C} \text{ is contained in } I\}$. (I1) is (C1); (I2) is immediate. For (I3) we induct on $|I \setminus J|$. Let I, J be $\mathcal{C}$ -independent with $|I| < |J|$.

If $I \setminus J = \emptyset$ then $I \subseteq J$, so any $j \in J \setminus I$ augments I . Now suppose $e \in I \setminus J$. The set $I \setminus \{e\}$ is independent of smaller excess over J , so induction supplies $f \in J \setminus I$ with $(I \setminus \{e\}) \cup \{f\}$ independent.

If $I \cup \{f\}$ is independent we are done. Otherwise the unique circuit $C \subseteq I \cup \{f\}$ contains both e and f (neither I nor $(I \setminus \{e\}) \cup \{f\}$ contains a circuit). Set $I' := (I \setminus \{e\}) \cup \{f\}$. Then $|I'| = |I| < |J|$ and $|I' \setminus J| = |I \setminus J| - 1$, so induction supplies $g \in J \setminus I'$ with $I' \cup \{g\}$ independent.

The element g also augments I . If not, a circuit $C' \subseteq I \cup \{g\}$ contains g and e (since $I' \cup \{g\}$ is independent). Then $e \in C \cap C'$, so (C3) yields a circuit contained in $(C \cup C') \setminus \{e\} \subseteq I' \cup \{g\}$, contradicting independence of $I' \cup \{g\}$. Finally $g \in J \setminus I$, as required.

Thus $(E, \mathcal{I})$ is a matroid whose circuits are exactly $\mathcal{C}$.

The weak elimination axiom (C3) has a stronger form, equally characteristic of matroids. **Strong circuit elimination:** if C_1, C_2 are distinct circuits, $e \in C_1 \cap C_2$, and $f \in C_1 \setminus C_2$, then some circuit C_3 satisfies $f \in C_3 \subseteq (C_1 \cup C_2) \setminus \{e\}$. (In graphs: the third cycle through a prescribed edge of $C_1 \setminus C_2$.) One also has uniqueness of the circuit in $I \cup \{e\}$ whenever I is independent and $I \cup \{e\}$ is dependent; this is the fundamental-circuit lemma of the next section.

A set X is spanning if and only if it meets every cocircuit (circuit of M^*). Dually, X is independent if and only if it meets no circuit. This is the first instance of the independence / spanning duality that organises the rest of the notes.

Given a basis B , the fundamental circuits $C(e, B)$ for $e \notin B$ already determine M : a set is dependent if and only if it contains one of these circuits or fails to be contained in a basis obtained by the usual exchanges. Equivalently, the fundamental cocircuits $C^*(b, E \setminus B)$ for $b \in B$ determine the dual, hence M .

6 Duality

The viewpoints $\mathcal{I}, \mathcal{B}, \mathcal{C}$ are not isolated; they complement one another. Duality is the first illustration.

6.1 The dual matroid

Theorem 6.1 (Dual matroid). *Let $M = (E, \mathcal{B})$ be a matroid. Set*

$$\mathcal{B}^* = \{E \setminus B : B \in \mathcal{B}\}.$$

Then $M^ = (E, \mathcal{B}^*)$ is a matroid, the **dual** of M .*



Example 6.2. For the running example $\mathcal{B} = \{abc, abd, abe, acd, ace\}$ one has

$$\mathcal{B}^* = \{def, cef, cdf, bef, bdf\}.$$

In the graphic picture, M^* is the matroid of the “dual” drawing in which the loop f becomes an isthmus and the parallel pair $\{d, e\}$ becomes a series pair. (In general the dual of a graphic matroid need not be graphic; it is planar-graphic precisely when G is planar.)



The proof proceeds as: a lemma on circuits, a lemma on bases, then the theorem.

Lemma 6.3 (Fundamental circuit). *If I is independent and $I \cup \{e\}$ is dependent, then $I \cup \{e\}$ contains a unique circuit, and that circuit contains e . It is denoted $C(e, I)$ and called the **fundamental circuit** of e with respect to I .*



Proof. The set $I \cup \{e\}$ is dependent, so it contains some circuit C ; $C \not\subseteq I$, hence $e \in C$. If C' were another circuit in $I \cup \{e\}$ through e , then (C3) would produce a circuit contained in $(C \cup C') \setminus \{e\} \subseteq I$, contradicting independence of I . $\square$

Lemma 6.4 (Symmetric exchange). *($B2^*$). If $A, B \in \mathcal{B}$ and $b \in B \setminus A$, then there exists $a \in A \setminus B$ such that $(A \setminus \{a\}) \cup \{b\} \in \mathcal{B}$.*



Proof. A is independent and $A \cup \{b\}$ is dependent (else A was not a basis), so $C = C(b, A)$ exists. The only circuit in $A \cup \{b\}$ is C , so $(A \setminus \{a\}) \cup \{b\}$ is independent — hence a basis — if and only if $a \in C$ and $a \neq b$. We need such an a in $A \setminus B$. Since B is independent and C is dependent, $C \not\subseteq B$, so there is $a \in C \setminus B \subseteq A \setminus B$. $\square$

Proof of Theorem 6.1. We check that $\mathcal{B}^*$ satisfies (B1)–(B2). Nonemptiness is immediate. For exchange, let $A^*, B^* \in \mathcal{B}^*$ and $a \in A^* \setminus B^*$. Writing $A^* = E \setminus A$ and $B^* = E \setminus B$ we have $A, B \in \mathcal{B}$ and $a \in B \setminus A$. Lemma 6.4 supplies $b \in A \setminus B$ with $(A \setminus \{b\}) \cup \{a\} \in \mathcal{B}$, which rearranges to $(A^* \setminus \{a\}) \cup \{b\} \in \mathcal{B}^*$. $\square$



Thus $(E, \mathcal{B})$ is a matroid if and only if $\mathcal{B}$ satisfies (B1) and ($B2^$). Indeed, M is a matroid if and only if M^* is, and basis exchange for M^* is symmetric exchange for M . See Oxley, Corollary 2.1.5.*

6.2 Linear duality

Let $V \subseteq \mathbb{F}^n$ be a k -dimensional subspace, with the standard coordinates. If the rows of a $k \times n$ matrix A span V , then the column matroid of A depends only on V ; write it $M(V)$.

Proposition 6.5. *The bases of $M(V)$ are the k -subsets $S \subseteq E = \{1, \dots, n\}$ for which $V \cap X_{E \setminus S} = \{0\}$, where $X_A = \{x \in \mathbb{F}^n : x_i = 0 \text{ for all } i \notin A\}$ is the coordinate subspace on A .*



Proof. The set S is a basis of the column matroid if and only if the corresponding k columns have rank k , if and only if the projection $\pi_S : \mathbb{F}^n \rightarrow X_S$ restricts to an injection on V . That happens if and only if $V \cap \ker \pi_S = \{0\}$. But $\ker \pi_S = X_{E \setminus S}$. $\square$

Theorem 6.6 (Orthogonal complements are duals). *If M is the matroid of a subspace $V \subseteq \mathbb{F}^n$, then M^* is the matroid of the orthogonal complement $V^\perp$. In particular, the dual of a linear matroid is linear.*



Proof. We use the identity $(U + W)^\perp = U^\perp \cap W^\perp$, whose proof is: $x \perp U + W$ if and only if $x \perp U$ and $x \perp W$. A set S is a basis of $M(V^\perp)$ if and only if $V^\perp \cap X_{E \setminus S} = \{0\}$, if and only if $V^\perp \cap X_S^\perp = \{0\}$ (since $X_{E \setminus S} = (X_S)^\perp$), if and only if $(V + X_S)^\perp = \{0\}$, if and only if $V + X_S = \mathbb{F}^n$, if and only if $V \cap X_S = \{0\}$, if and only if $E \setminus S$ is a basis of $M(V)$, if and only if S is a basis of $M(V)^*$. $\square$

 A generic k -plane meets a generic $(n - k)$ -plane trivially, so a generic k -plane V has every k -set as a basis: this is the **uniform matroid** $U_{k,n}$. The matroid $M(V)$ records how special V is with respect to the coordinate subspaces.

6.3 Graphic duality

Think of vertices as points and edges as continuous curves.

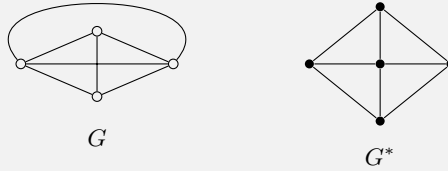
Definition 6.7. A graph is **planar** if it can be drawn in the plane with no two edges meeting except at a common endpoint. A **plane graph** is a graph already so drawn. The drawing divides the plane into regions, the **faces**. 

Definition 6.8 (Dual plane graph). The dual G^* of a plane graph G is obtained by placing a vertex f^* in each face f of G , and, for each edge e of G incident to faces f and g , drawing an edge e^* from f^* to g^* that crosses e and no other edge of G . 

Observation 6.9. If G is connected then $(G^*)^* \cong G$. 

Loops in G become isthmuses in G^* and conversely; a cycle in G becomes a cut in G^* .

Example 6.10. A typical plane graph and its dual (the correspondence is $V \leftrightarrow F^*$, $E \leftrightarrow E^*$, $F \leftrightarrow V^*$):

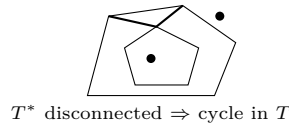


Theorem 6.11. If G is a connected plane graph and G^* is its dual, then $M(G^*) = M(G)^*$. 

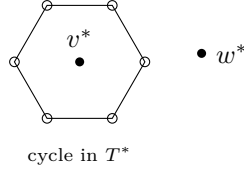
Proof. It is enough to show that $T \subseteq E$ is a spanning tree of G if and only if $E \setminus T$ (viewed as edges of G^*) is a spanning tree of G^* . By duality it suffices to prove one implication and apply it to G^* .

Suppose T is a spanning tree of G , and write $T^* = E \setminus T$ as a set of edges of G^* .

If T^* were disconnected, two vertices f^*, g^* of G^* would lie in different components of $(V(G^*), T^*)$. The corresponding faces f, g of G would then be separated by a cut made of edges of T , forcing a cycle in T , a contradiction.



If T^* contained a cycle C^* , take a vertex v of G inside C^* and a vertex w outside it. Any v – w path in G must cross C^* , hence use an edge of T^* , i.e. an edge not in T . Thus v and w are disconnected in T , contradicting that T is spanning.



□

Corollary 6.12 (Euler's formula). *For a connected plane graph, $|V| - |E| + |F| = 2$.*

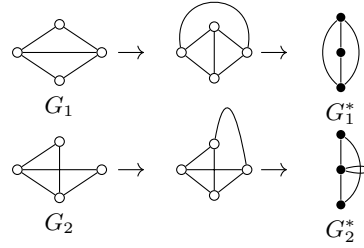


Proof. A spanning tree T of G has $|V| - 1$ edges. The complementary spanning tree T^* of G^* has $|F| - 1$ edges (including the unbounded face). But $T^* = E \setminus T$, so

$$|F| - 1 = |E| - (|V| - 1),$$

which rearranges to Euler's formula. □

Observation 6.13. *Different plane embeddings of the same abstract graph may produce non-isomorphic dual graphs $G_1^* \not\cong G_2^*$. Nevertheless $M(G_1^*) = M(G_2^*) = M(G)^*$.*



(G_1^* has a vertex of degree 6; G_2^* does not.) Natural projects: (i) how are the various duals of G related? (ii) when is $M(G) \cong M(H)$?

Duality is an involution: $(M^*)^* = M$, because taking complements twice recovers every basis. The dual rank formula

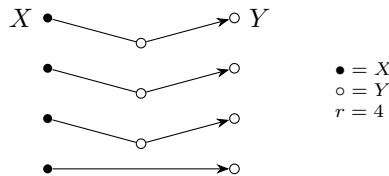
$$r_{M^*}(X) = |X| - r(M) + r(E \setminus X)$$

is recorded with a proof in the minors chapter; it is the numerical content of “bases of M^* are complements of bases of M ”. Loops of M are coloops of M^* , and circuits of M are cocircuits of M^* .

Whitney's planarity criterion is the global form of Theorem 6.11: an abstract graph G is planar if and only if $M(G)^*$ is graphic. Combined with Tutte's excluded-minor theorem for graphic matroids, this recovers Kuratowski–Wagner (K_5 and $K_{3,3}$) as the statement that $M(K_5)^*$ and $M(K_{3,3})^*$ are the extra forbidden minors beyond $U_{2,4}$, F_7 , and F_7^* .

6.4 Transversal duality

Definition 6.14 (Routing). *Let G be a directed graph and $X, Y \subseteq V(G)$ with $|X| = |Y| = r$. A **routing** from X to Y is a set of r directed paths, vertex-disjoint, starting in X and ending in Y . (A path may be a single vertex.)*



Theorem 6.15 (Mason, 1972). Let $G = (V, E)$ be a digraph and $B_0 \subseteq V$. The collection

$$L(G, B_0) = \{X \subseteq V : \text{there is a routing from } X \text{ to } B_0\}$$

is the set of bases of a matroid on V . Such matroids are called *cotransversal* (or *strict gammoids*). 

The board notes state Mason's theorem without a proof. Independence of a set X can be read as the existence of $|X|$ vertex-disjoint directed paths from X into B_0 ; the augmentation axiom is then Menger's theorem in the digraph obtained by attaching a source to X and a sink to B_0 .

Theorem 6.16 (Ingleton–Piff, 1973). Cotransversal matroids are precisely the duals of transversal matroids. 

The notes defer this to Homework 3. A later sketch (Ardila, 2006) runs: transversal matroids are linear, represented by row-spaces $V \subseteq \mathbb{R}^n$; cotransversal matroids are linear, represented by subspaces $W \subseteq \mathbb{R}^n$; and the subspaces arising in the two constructions are orthogonal complements. Combined with Theorem 6.6 this yields Ingleton–Piff.

7 Rank and closure

7.1 Rank

Definition 7.1 (Rank function). The *rank* of $A \subseteq E$ is

$$r(A) = \max\{|I| : I \subseteq A, I \in \mathcal{I}\},$$

the size of any maximal independent subset of A . Homework 2 in the notes records that $(A, \mathcal{I} \cap 2^A)$ is itself a matroid (the *restriction* $M|A$, written $M \setminus (E \setminus A)$ later); all its bases have the same size by Proposition 3.2, so the maximum is attained and is independent of the choice of basis. The rank of M is $r(M) = r(E)$. 

Proposition 7.2. The rank function satisfies, for all $X, Y \subseteq E$,

- (R1) $0 \leq r(X) \leq |X|$;
 - (R2) $X \subseteq Y$ implies $r(X) \leq r(Y)$;
 - (R3) $r(X \cup Y) + r(X \cap Y) \leq r(X) + r(Y)$ (submodularity).
- 

Proof. (R1) and (R2) are immediate. For (R3), let A be a basis of $X \cap Y$, and extend A to a basis B of $X \cup Y$. Write $B = A \sqcup C \sqcup D$ with $C \subseteq X \setminus Y$ and $D \subseteq Y \setminus X$ (elements of $B \setminus A$ lying in $X \cap Y$ cannot occur: they would enlarge the basis A of $X \cap Y$). Then

$$r(X \cup Y) = |B| = |A| + |C| + |D|, \quad r(X \cap Y) = |A|,$$

while $A \cup C \subseteq X$ is independent so $r(X) \geq |A| + |C|$, and likewise $r(Y) \geq |A| + |D|$. Adding these inequalities gives (R3). $\square$

Lemma 7.3. If $r(X \cup \{y\}) = r(X)$ for every $y \in Y$, then $r(X \cup Y) = r(X)$. 

Proof. Induct on $|Y|$. The cases $|Y| \leq 1$ are tautological. Write $Y = Y' \cup \{y\}$ with $|Y'| = n - 1$. The inductive hypothesis gives $r(X \cup Y') = r(X)$. Submodularity then yields

$$\begin{aligned} r(X \cup Y) &= r((X \cup Y') \cup (X \cup \{y\})) \\ &\leq r(X \cup Y') + r(X \cup \{y\}) - r(X) = r(X) + r(X) - r(X) = r(X), \end{aligned}$$

and the reverse inequality is (R2). $\square$

Theorem 7.4. *A function $r: 2^E \rightarrow \mathbb{N}$ is the rank function of a matroid on E if and only if it satisfies (R1)–(R3).*



Proof. The forward direction is Proposition 7.2. Conversely, set $\mathcal{I} = \{I \subseteq E : r(I) = |I|\}$. We check that $(E, \mathcal{I})$ is a matroid whose rank function is the given r .

(I1): (R1) forces $r(\emptyset) = 0 = |\emptyset|$. (I2): if $I \subseteq J$ and $r(J) = |J|$, submodularity gives $r(I) + r(J \setminus I) \geq r(J) + r(\emptyset) = |J|$, while $r(J \setminus I) \leq |J \setminus I|$, so $r(I) \geq |I|$. Combined with (R1) we get $r(I) = |I|$.

For (I3), let $I, J \in \mathcal{I}$ with $|I| < |J|$, and suppose $I \cup \{j\} \notin \mathcal{I}$ for every $j \in J \setminus I$. Then $r(I \cup \{j\}) < |I| + 1$. But $r(I \cup \{j\}) \geq r(I) = |I|$, so $r(I \cup \{j\}) = r(I)$. Lemma 7.3 yields $r(I \cup J) = r(I) = |I| < |J| = r(J) \leq r(I \cup J)$, a contradiction.

Finally, the rank of this matroid recovers r . Let $X \subseteq E$ and let $I \subseteq X$ be a maximal member of $\mathcal{I}$. Then $r(I) = |I|$. For each $y \in X \setminus I$ the set $I \cup \{y\}$ is not independent, so $r(I \cup \{y\}) = |I| = r(I)$. Lemma 7.3 gives $r(X) = r(I) = |I|$, which is the matroid rank of X . $\square$

7.2 Closure

Definition 7.5 (Closure). *The **closure** of $X \subseteq E$ is*

$$\text{cl}(X) = \{x \in E : r(X \cup \{x\}) = r(X)\}.$$



This is “span, for matroids”. It satisfies the Kuratowski closure axioms, plus the MacLane–Steinitz exchange axiom:

(CL1) $X \subseteq \text{cl}(X)$;

(CL2) $X \subseteq Y$ implies $\text{cl}(X) \subseteq \text{cl}(Y)$;

(CL3) $\text{cl}(\text{cl}(X)) = \text{cl}(X)$;

(CL4) if $x \in \text{cl}(X \cup \{y\})$ and $x \notin \text{cl}(X)$, then $y \in \text{cl}(X \cup \{x\})$.

Analogues: topological closure, linear span, the vertices generating a connected subgraph.

Lemma 7.6. *For any $X \subseteq E$ and $x \in E$, either $r(X \cup \{x\}) = r(X)$ or $r(X \cup \{x\}) = r(X) + 1$.*



Proof. Let B be a basis of X . Then either B or $B \cup \{x\}$ is a basis of $X \cup \{x\}$. $\square$

Proof of the closure axioms. (CL1) is Lemma 7.6 together with $r(X \cup \{x\}) = r(X)$ when $x \in X$.

(CL2). Let $X \subseteq Y$ and $x \in \text{cl}(X)$, so $r(X \cup \{x\}) = r(X)$. Let B be a basis of X ; then B is also a basis of $X \cup \{x\}$, i.e. $B \cup \{x\}$ is dependent. Extend B to a basis C of $Y \cup \{x\}$. Then $x \notin C$ (else $B \cup \{x\}$ would be independent), so $C \subseteq Y$ and $r(Y) \geq |C| = r(Y \cup \{x\})$. Thus $r(Y) = r(Y \cup \{x\})$ and $x \in \text{cl}(Y)$.

(CL3). Always $X \subseteq \text{cl}(X) \subseteq \text{cl}(\text{cl}(X))$. If $x \in \text{cl}(\text{cl}(X))$ then $r(\text{cl}(X) \cup \{x\}) = r(\text{cl}(X))$. The local rank lemma Lemma 7.3 gives $r(\text{cl}(X)) = r(X)$, because $r(X \cup \{y\}) = r(X)$ for every $y \in \text{cl}(X)$. Hence $r(X \cup \{x\}) = r(X)$ and $x \in \text{cl}(X)$.

(CL4). Suppose $y \in \text{cl}(X \cup \{x\})$ and $y \notin \text{cl}(X)$. Then $r(X \cup \{x, y\}) = r(X \cup \{x\})$ and $r(X \cup \{y\}) = r(X) + 1$. Therefore

$$r(X \cup \{x, y\}) = r(X \cup \{x\}) \leq r(X) + 1 = r(X \cup \{y\}),$$

so $r(X \cup \{x\}) = r(X \cup \{y\})$ and $x \in \text{cl}(X \cup \{y\})$. $\square$

Theorem 7.7. *A map $\text{cl}: 2^E \rightarrow 2^E$ is the closure operator of a matroid on E if and only if it satisfies (CL1)–(CL4).*



The converse. Given cl , declare I independent if $x \notin \text{cl}(I \setminus \{x\})$ for every $x \in I$.

(I1) is vacuous. (I2): if $I \subseteq J$ with J independent and $x \in I$, then $I \setminus \{x\} \subseteq J \setminus \{x\}$, so $\text{cl}(I \setminus \{x\}) \subseteq \text{cl}(J \setminus \{x\})$ by (CL2), hence $x \notin \text{cl}(J \setminus \{x\})$ forces $x \notin \text{cl}(I \setminus \{x\})$.

If I is independent and $e \notin I$, then $I \cup \{e\}$ is dependent if and only if $e \in \text{cl}(I)$.

If $e \in \text{cl}(I)$, then $e \in \text{cl}((I \cup \{e\}) \setminus \{e\})$, so $I \cup \{e\}$ is dependent. Conversely, if $I \cup \{e\}$ is dependent then some $x \in I \cup \{e\}$ satisfies $x \in \text{cl}((I \cup \{e\}) \setminus \{x\})$. If $x = e$ we are done. If $x \in I$, then $x \in \text{cl}((I \setminus \{x\}) \cup \{e\})$ while $x \notin \text{cl}(I \setminus \{x\})$ (since I is independent). (CL4) yields $e \in \text{cl}((I \setminus \{x\}) \cup \{x\}) = \text{cl}(I)$.

For (I3), let $I, J \in \mathcal{I}$ with $|I| < |J|$, and suppose $I \cup \{j\}$ is dependent for every $j \in J \setminus I$. The lemma gives $J \setminus I \subseteq \text{cl}(I)$, hence $J \subseteq \text{cl}(I)$ by (CL1). Independence of J says that each $y \in J$ satisfies

$y \notin \text{cl}(J \setminus \{y\})$. But $J \setminus \{y\} \subseteq \text{cl}(I)$, so $\text{cl}(J \setminus \{y\}) \subseteq \text{cl}(\text{cl}(I)) = \text{cl}(I)$ by (CL2)–(CL3), and therefore $y \notin \text{cl}(I)$. This forces $J = \emptyset$, contradicting $|J| > |I|$.

The closure of the resulting matroid is the given operator: e is in the matroid closure of X if and only if e does not enlarge a basis of X , if and only if $e \in \text{cl}(X)$ by the lemma (applied after extending a basis of X using (CL3)). $\square$

 A closure operator with the exchange property (CL4) is a matroid (combinatorial independence). A closure operator with the anti-exchange property

$$x \in \text{cl}(X \cup \{y\}) \setminus \text{cl}(X) \implies y \notin \text{cl}(X \cup \{x\})$$

is a convex geometry (combinatorial convexity). The convex hull is the motivating example of anti-exchange.

Concretely: if x lies in the convex hull of $X \cup \{y\}$ but not of X , then y cannot lie in the convex hull of $X \cup \{x\}$ (else x and y would be interchangeable, as in a linear dependence). Three collinear points in the plane give a convex geometry that is not a matroid, because the middle point is in the hull of the outer two, violating exchange.

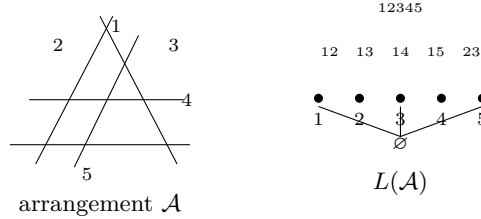
Two more families of sets, dual to independent sets and circuits. A set X is *spanning* if $r(X) = r(E)$, equivalently $\text{cl}(X) = E$. The minimal spanning sets are exactly the bases. A *hyperplane* is a maximal non-spanning set, equivalently a flat of rank $r(M) - 1$; the hyperplanes of M are the complements of the cocircuits. Thus: independent $\leftrightarrow$ co-spanning, circuit $\leftrightarrow$ cocircuit, spanning $\leftrightarrow$ co-independent, hyperplane $\leftrightarrow$ complement of a circuit of M^* .

A set I is independent if and only if $r(I) = |I|$, and X is spanning if and only if $r(X) = r(E)$. Combining both: B is a basis if and only if $r(B) = |B| = r(E)$.

8 Flats and geometric lattices

8.1 Motivation: hyperplane arrangements

A real hyperplane arrangement $\mathcal{A}$ has an *intersection poset* $L(\mathcal{A})$, whose elements are the intersections of subfamilies of $\mathcal{A}$, ordered by reverse inclusion.



Order: $x \leq y$ in $L(\mathcal{A})$ if $x \supseteq y$. This is a poset. How much of the geometry of $\mathcal{A}$ is recorded by $L(\mathcal{A})$ (number of regions, bounded regions, topology)? A great deal. The next construction generalises this: from a matroid M one builds the *lattice of flats* $L(M)$.

8.2 Flats

Definition 8.1 (Flat). A *flat* of M is a set $F \subseteq E$ with $\text{cl}(F) = F$. Equivalently, $r(F \cup \{x\}) = r(F) + 1$ for every $x \notin F$.

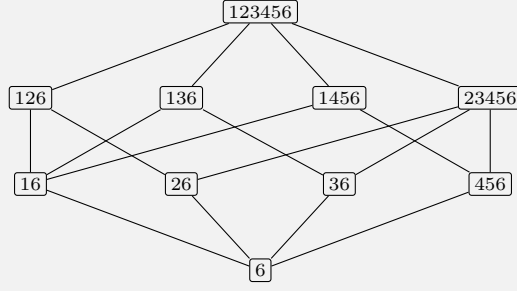


Example 8.2. If M is represented by a vector configuration, flats correspond to intersections of the configuration with linear subspaces. In the running example (labelling $1, \dots, 6$) the flats are

$$\{6\}; \quad \{1, 6\}, \{2, 6\}, \{3, 6\}, \{4, 5, 6\}; \quad \{1, 2, 6\}, \{1, 3, 6\}, \{1, 4, 5, 6\}, \{2, 3, 4, 5, 6\}; \quad E.$$

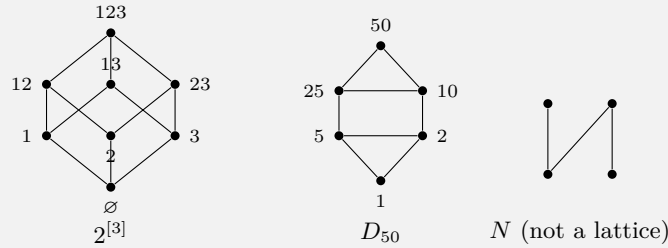
Ordered by inclusion they form the Hasse diagram below.





Definition 8.3 (Poset). A *partially ordered set* (poset) is a set P with a binary relation $\leq$ that is reflexive, antisymmetric, and transitive. For $a \neq b$ one may have $a < b$, $a > b$, or incomparability.

Example 8.4. Boolean lattice 2^S (by inclusion); divisor lattice D_n (by divisibility); $(\mathbb{Z}, \leq)$; the flats of M ordered by inclusion.



Say a *covers* b , written $a \succ b$, if $a > b$ and nothing lies strictly between them. The Hasse diagram of P has an edge for each cover.

Definition 8.5 (Lattice). A poset is a *lattice* if every pair of elements has a least upper bound (join $a \vee b$) and a greatest lower bound (meet $a \wedge b$).

The Boolean lattice, D_n , and $(\mathbb{Z}, \leq)$ are lattices; the N -shaped Hasse diagram is not.

8.3 The lattice of flats

Proposition 8.6. The flats of a matroid, ordered by inclusion, form a lattice $L(M)$.

Proof. Let F, G be flats. The meet is $F \wedge G = F \cap G$: the intersection of flats is a flat, because if $x \notin F \cap G$ then $x \notin F$ or $x \notin G$, say $x \notin F$, and submodularity gives

$$r(F) + r((F \cap G) \cup \{x\}) \geq r(F \cup \{x\}) + r(F \cap G) = r(F) + 1 + r(F \cap G),$$

so $r((F \cap G) \cup \{x\}) \geq r(F \cap G) + 1$. The join is $F \vee G = \text{cl}(F \cup G)$, which is the smallest flat containing both. $\square$

Definition 8.7. A poset is *graded* if it admits a rank function $r: P \rightarrow \mathbb{N}$ with $r = 0$ on minimal elements and $r(y) = r(x) + 1$ whenever y covers x .

Proposition 8.8. $L(M)$ is graded by the matroid rank. The unique minimal element is $\text{cl}(\emptyset)$ (the set of loops), of rank 0. If $F \leq G$ and $x \in G \setminus F$, then $F \subsetneq \text{cl}(F \cup \{x\}) \subseteq G$ and $r(\text{cl}(F \cup \{x\})) = r(F) + 1$, so $G = \text{cl}(F \cup \{x\})$.

Proposition 8.9 (Semimodularity). For flats F, G , $r(F) + r(G) \geq r(F \vee G) + r(F \wedge G)$.

Proof. Submodularity of matroid rank, together with $r(\text{cl}(F \cup G)) = r(F \cup G)$. $\square$

Proposition 8.10 (Atomic). *Every flat is a join of rank-1 flats (atoms).* 

Proof. For $F \in L(M)$ and $f \in F$ the singleton closure $\text{cl}(\{f\})$ is an atom contained in F (or f is a loop, hence already in $\text{cl}(\emptyset)$). Then $F = \text{cl}(\bigcup_{f \in F} \text{cl}(\{f\})) = \bigvee_{f \in F} \text{cl}(\{f\})$. $\square$

Finite meets and joins exist in any finite lattice: $F \wedge G \wedge \dots = F \cap G \cap \dots$ and $F \vee G \vee \dots = \text{cl}(F \cup G \cup \dots)$ in $L(M)$.

8.4 Geometric lattices

Definition 8.11. A finite lattice is *geometric* if it is semimodular and atomic. (Semimodularity implies graded.) 

Theorem 8.12. A finite lattice is geometric if and only if it is the lattice of flats of a matroid. 

Proof. We have already seen $L(M)$ is geometric. Conversely, let L be geometric and let E be its set of atoms (or $E = \emptyset$ if L is a point). Define $r: 2^E \rightarrow \mathbb{N}$ by

$$r(\{e_1, \dots, e_k\}) = r_L(e_1 \vee \dots \vee e_k).$$

This satisfies the local rank axioms

(R1') $r(\emptyset) = 0$;

(R2') $r(A \cup \{a\}) - r(A) \in \{0, 1\}$;

(R3') if $r(A) = r(A \cup \{a\}) = r(A \cup \{b\})$ then $r(A \cup \{a, b\}) = r(A)$.

Indeed (R1') is $r_L(\hat{0}) = 0$. For (R2'), semimodularity gives

$$0 \leq r_L(x \vee a) - r_L(x) \leq r_L(a) - r_L(x \wedge a) \leq 1$$

with $x = \bigvee A$. For (R3'), if $r(A) = r(A \cup \{a\}) = r(A \cup \{b\})$ then $x = x \vee a = x \vee b$, so $x \vee a \vee b = x$.

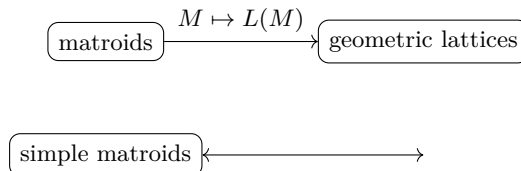
These local axioms imply (R1)–(R3), so r is a matroid rank function. Indeed: $r(\emptyset) = 0$ is (R1'), and iterating (R2') gives $0 \leq r(A) \leq |A|$ and monotonicity. For submodularity, add the elements of $Y \setminus X$ one at a time; whenever an element does not increase rank, (R3') says it remains redundant after any further additions, which is the local rank lemma and hence (R3). The map

$$\phi: L \rightarrow L(M), \quad x \mapsto \{\text{atoms } a \leq x\}$$

is a poset isomorphism: it is well-defined (the image is a flat, else semimodularity would force a missing atom under x), injective because L is atomic (so $x = \bigvee \{a : a \leq x\}$), surjective by construction of r , and order-preserving in both directions. $\square$

8.5 Simple matroids

The correspondence $M \mapsto L(M)$ loses loops and parallel elements, and nothing else.



Definition 8.13. A matroid is *simple* (a combinatorial geometry) if it has no loops and no parallel elements. The *simplification* of M is obtained by deleting loops and keeping one element from each parallel class. 

Proposition 8.14. *Simple matroids are in bijection with geometric lattices.*



Proposition 8.15. *Geometric lattices are coatomic: every flat of corank k is a meet of k coatoms.*



Proof. Coatoms of $L(M)$ are the rank- $(r(M)-1)$ flats, i.e. the **hyperplanes** (maximal non-spanning sets). Induct on k . Let X be a flat of rank $r-k$, pick $y \notin X$, and set $Y = \text{cl}(X \cup \{y\})$, so Y covers X and $r(Y) = r-k+1$. **The board drawing of the two cases is unlabeled; the split is:** if $E \setminus \{y\}$ is non-spanning let H_Y be a hyperplane containing it and X ; if $E \setminus \{y\}$ is spanning, enlarge X inside $E \setminus \{y\}$ to a hyperplane H_Y . Either way $X = Y \wedge H_Y$ with $Y \not\leq H_Y$. Apply induction to Y . $\square$

The several cryptomorphisms of the notes may be summarised as a single dictionary. Each column determines the others.

data	recovers independence by
independent sets $\mathcal{I}$	given
bases $\mathcal{B}$	subsets of members of $\mathcal{B}$
circuits $\mathcal{C}$	sets containing no member of $\mathcal{C}$
rank r	$r(I) = I $
closure cl	$x \notin \text{cl}(I \setminus \{x\})$ for all $x \in I$
flats $L(M)$	r on the lattice, after adding loops/parallels

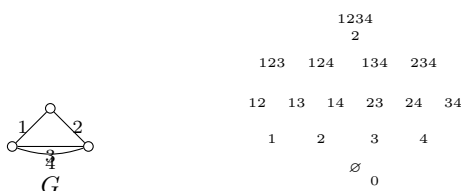
The missing direction from $L(M)$ back to M is the simplification discussed above: the lattice remembers the simple matroid, and one restores loops and parallel classes by hand. Duality reads, on this dictionary, as independent $\leftrightarrow$ spanning in the dual, circuit $\leftrightarrow$ cocircuit, flat $\leftrightarrow$ cyclic flat of the dual, and $r^*(X) = |X| - r(E) + r(E \setminus X)$.

9 Minors

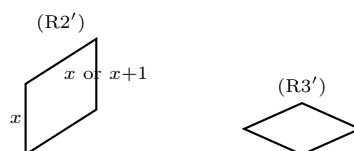
Vector spaces have subspaces; graphs have subgraphs. Matroids have **minors**.

9.1 Rank on the Boolean lattice

One way to think about a matroid is as a labelling of 2^E by integers. For the triangle G with edges $\{1, 2, 3, 4\}$ as drawn (bases $\{12, 13, 14, 23, 24\}$):



A rank function is a labelling of 2^E such that along any covering $X \subset X \cup \{e\}$ the label stays or increases by 1, and every “parallelogram” $X, X \cup \{a\}, X \cup \{b\}, X \cup \{a, b\}$ is submodular. Subcubes inherit these properties, so they are matroids: this is deletion and contraction.



9.2 Deletion and contraction

Definition 9.1 (Deletion). *For $T \subseteq E$, the **deletion** $M \setminus T$ (restriction $M|(E \setminus T)$) has ground set $E \setminus T$ and rank $r_{M \setminus T}(X) = r_M(X)$ for $X \subseteq E \setminus T$.*



Definition 9.2 (Contraction). The *contraction* M/T has ground set $E \setminus T$ and rank

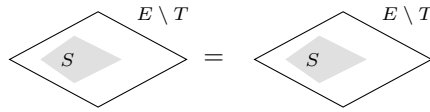
$$r_{M/T}(X) = r_M(X \cup T) - r_M(T).$$



Proposition 9.3. If $S \cap T = \emptyset$ then

$$\begin{aligned} (M \setminus S) \setminus T &= M \setminus (S \cup T) = (M \setminus T) \setminus S, \\ (M/S)/T &= M/(S \cup T) = (M/T)/S, \\ (M/S) \setminus T &= (M \setminus T)/S. \end{aligned}$$

Thus deletion and contraction commute: a string of deletions and contractions may be rewritten in any order. For instance $M/2 \setminus 59/6/7 \setminus 8 = M/267 \setminus 589$.



Definition 9.4 (Minor). A *minor* of M is a matroid $M/S \setminus T$ with $S \cap T = \emptyset$.

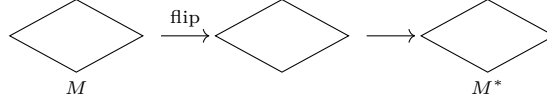


Proposition 9.5 (Duality interchanges deletion and contraction).

$$(M/T)^* = M^* \setminus T, \quad (M \setminus T)^* = M^*/T.$$



Proposition 9.6. The dual rank is $r_{M^*}(X) = |X| - r_M(E) + r_M(E \setminus X)$.



This is Homework 4 in the notes; the board “flip the Boolean lattice” picture is unlabeled. In symbols: a basis of M^* contained in X is $X \setminus B$, where B is a basis of M that meets $E \setminus X$ in as many elements as possible, i.e. $|B \cap (E \setminus X)| = r(E \setminus X)$. Then

$$r_{M^*}(X) = |X| - (r(M) - r(E \setminus X)) = |X| - r(M) + r(E \setminus X).$$

Proposition 9.7.

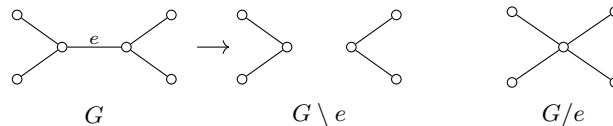
$$\begin{aligned} \mathcal{I}(M \setminus T) &= \{I \subseteq E \setminus T : I \in \mathcal{I}(M)\}, \\ \mathcal{I}(M/T) &= \{I \subseteq E \setminus T : I \cup B_T \in \mathcal{I}(M)\}, \end{aligned}$$

where B_T is any basis of T . Circuits: $\mathcal{C}(M \setminus T) = \{C \in \mathcal{C}(M) : C \cap T = \emptyset\}$, while $\mathcal{C}(M/T)$ consists of the inclusion-minimal sets among $\{C \setminus T : C \in \mathcal{C}(M)\}$. Closure in the contraction: $\text{cl}_{M/T}(X) = \text{cl}_M(X \cup T) \setminus T$.



9.3 Graphic minors

Graph deletion of an edge e is ordinary deletion of e . Graph contraction of $e = i_1 i_2$ identifies i_1 with i_2 .



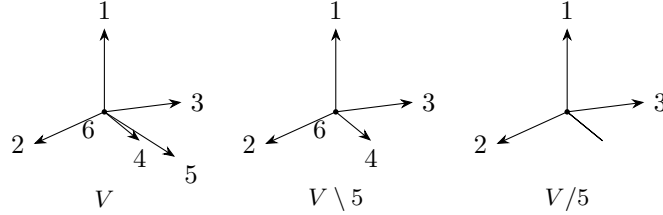
Proposition 9.8. $M(G \setminus e) = M(G) \setminus e$ and $M(G/e) = M(G)/e$. Consequently every minor of a graphic matroid is graphic.



Proof. An edge-set $I \not\ni e$ is acyclic in $G \setminus e$ if and only if it is acyclic in G . If e is not a loop, I is independent in $M(G)/e$ if and only if $I \cup \{e\}$ is acyclic in G , if and only if I is acyclic in G/e . $\square$

9.4 Linear minors

Deleting a vector e from a configuration $E \subseteq V$ yields $E \setminus \{e\}$ in V . Contracting e is the configuration $\pi(E \setminus \{e\})$ in $e^\perp$, where $\pi: V \rightarrow e^\perp$ is orthogonal projection.



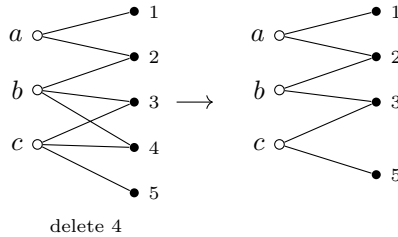
Proposition 9.9. $M(E \setminus e) = M(E) \setminus e$ and $M(E/e) = M(E)/e$. Every minor of a linear matroid is linear.



Proof. The deletion claim is immediate. For contraction: I is independent in $M(E)/e$ if and only if $I \cup \{e\}$ is linearly independent, if and only if there is no linear relation among $\pi(I)$, because a relation $\sum c_i v_i + ce = 0$ projects to $\sum c_i \pi(v_i) = 0$, and conversely a relation among the projections lifts by an element of $\ker \pi = \langle e \rangle$. $\square$

9.5 Transversal minors and gammoids

Is every minor of a transversal matroid transversal? Deletion is: if e “decides not to marry”, erase e from the bipartite graph.

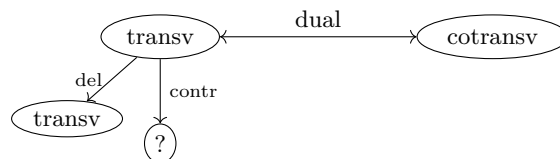


Contraction is not: forcing e to be matched can produce a non-transversal matroid.

Example 9.10. A transversal matroid $M = M(G)$ with a non-transversal contraction M/a . The original drawing is cramped; the intended configuration is: the circuits of M include $\{b_1, b_2\}$, $\{c_1, c_2\}$, $\{d_1, d_2\}$ and $\{a, b_i\}$, $\{a, c_j\}$, $\{a, d_k\}$. After contracting a one obtains circuits $\{b_1, b_2\}$, $\{c_1, c_2\}$, $\{d_1, d_2\}$ together with all mixed triples $b_i c_j d_k$, which no bipartite presentation realises (any presentation would have to match the three pairs from three distinct men, yet also match a mixed triple, forcing a Hall violator).



Graphic and linear matroids are closed under minors and duality. Transversal matroids are closed under deletion and dualise to cotransversal matroids, but are not closed under contraction.



Definition 9.11 (Gammoid). A *gammoid* is a contraction of a transversal matroid.



Proposition 9.12. Gammoids are closed under minors and duality, and they form the smallest such class containing the transversal matroids.



Proof. Let $M = T/B$ with T transversal. Contraction: $M/A = T/(A \cup B)$, a gammoid. Deletion: $M \setminus A = (T \setminus A)/B$, and $T \setminus A$ is transversal. Duality: $T = U \setminus C$ for a cotransversal U (Ingleton–Piff), so $M^* = (U \setminus C/B)^* = U^*/C \setminus B$ with U^* transversal, hence a gammoid. $\square$

The Tutte polynomial of the greedy chapter is most cleanly computed by the same two operations. If e is neither a loop nor a coloop,

$$T_M(x, y) = T_{M \setminus e}(x, y) + T_{M/e}(x, y);$$

a coloop contributes a factor x and a loop a factor y . In particular $T_M(1, 1)$ is the number of bases, $T_M(2, 0)$ is (for a graphic matroid) the number of acyclic orientations, and $T_M(0, 2)$ counts totally cyclic orientations. The same recurrence computes the chromatic polynomial of a graph as $\chi_G(\lambda) = (-1)^{r(M(G))} \lambda^{\kappa(G)} T_{M(G)}(1 - \lambda, 0)$.

A useful numerical check: in the running example $r(M) = 3$, there are five bases, a unique loop f , and a unique coloop a . Deleting f and contracting a leaves a 4-element rank-2 matroid (a triangle with one edge doubled), whose Tutte polynomial is $x^2 + x + y + xy + y^2$; multiplying by the loop and coloop factors y and x recovers T_M .

A minor of M can always be realised as a restriction of a contraction of a restriction (the *scum theorem*): after simplifying, every minor appears as a restriction of M/F for some flat F . In particular it is enough, when testing a minor-closed property, to look at restrictions of contractions.

Series and parallel connections of matroids generalise identifying endpoints of graphs. The cycle matroids of series-parallel graphs are precisely the graphic matroids with no $M(K_4)$ minor; they are the building blocks of regular matroids of small connectivity.

10 Forbidden minors and representability

Closure under minors and duality is useful because it yields characterisations by excluded minors.

	minors	duality
graphic	yes	no
linear	yes	yes
transversal	no	no
gammoid	yes	yes

10.1 Graphic matroids

Example 10.1. To test whether a list $\mathcal{C}$ is the circuit list of a graphic matroid, pass to a minor. If $\mathcal{C} = \{1279, 1289, 13456, 1379, 2345, 2567, 2789\}$, then

$$M \setminus \{3, 4, 5, 6\}/9$$

has circuits $\{127, 128, 178, 278\}$, i.e. is $U_{2,4}$, which is not graphic.



Proposition 10.2. $U_{m,n}$ is graphic only for the values $(m, n) \in \{(0, n), (1, n), (n-1, n), (n, n)\} \cup \{(2, n) : n \leq 3\}$. In particular $U_{2,4}$ is not graphic.



Proof. Deletion and contraction of uniform matroids stay uniform: $U_{m,n}/e \cong U_{m-1,n-1}$ and $U_{m,n} \setminus e \cong U_{m,n-1}$. Starting from $U_{m,n}$ with $2 \leq m \leq n-2$ one reaches $U_{2,4}$ as a minor, and $U_{2,4}$ is the cycle matroid of no graph (it would be a 2-connected graph on 4 edges with every 2-set independent, hence K_4 minus two edges, which has a 3-circuit). $\square$

Other non-graphic examples: $M(K_5)^*$, $M(K_{3,3})^*$, and the Fano matroid F_7 and its dual.

Theorem 10.3 (Tutte, 1959). *A matroid is graphic if and only if it has no minor isomorphic to $U_{2,4}$, $M(K_5)^*$, $M(K_{3,3})^*$, F_7 , or F_7^* .*



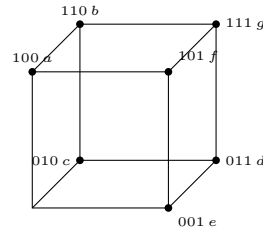
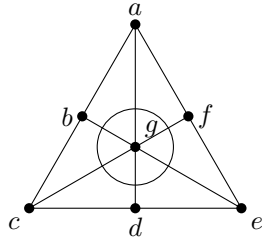
The proof is omitted on the board (it occupies Oxley, Chapter 13); the five matroids are exhibited as minors, but the converse—that every non-graphic matroid has one of them—is a long structural argument. These five are the *forbidden minors* for graphic matroids.

10.2 The Fano matroid

The Fano matroid F_7 has ground set $\{a, b, c, d, e, f, g\}$ and circuits the seven 3-point lines

$$\{abc, cde, efa, agd, bge, cgf, bdf\}.$$

It is the matroid of $\text{PG}(2, 2)$: the seven nonzero vectors in $\mathbb{F}_2^3$.



Proposition 10.4. *F_7 is not representable over $\mathbb{R}$ (nor over any field of characteristic not 2).*



Proof. Suppose it is represented in $\mathbb{R}^3$. We may take $a = (1, 0, 0)$, $c = (0, 1, 0)$, $e = (0, 0, 1)$, and $g = (x, y, z)$ with $xyz \neq 0$. The three lines through g force $d \parallel (0, y, z)$, $f \parallel (x, 0, z)$, $b \parallel (x, y, 0)$. The remaining line $\{b, d, f\}$ then gives a linear dependence $\alpha d + \beta f + \gamma b = 0$ whose three coordinates read $x(\beta + \gamma) = y(\alpha + \gamma) = z(\alpha + \beta) = 0$. Thus $\alpha + \beta = \beta + \gamma = \gamma + \alpha = 0$, so $2\alpha = 0$. Over $\mathbb{R}$ this forces $\alpha = 0$ and then $\beta = \gamma = 0$, a contradiction. Hence F_7 is not $\mathbb{R}$ -representable; the same computation in characteristic not 2 gives $2 = 0$, likewise a contradiction. $\square$

Corollary 10.5. *F_7^* is not representable over $\mathbb{R}$.*



If F_7^* had an $\mathbb{R}$ -representation, then Theorem 6.6 would give an $\mathbb{R}$ -representation of its dual F_7 , contradicting the proposition.

10.3 Representability over finite fields

Theorem 10.6 (Tutte, 1958). *A matroid is representable over $\text{GF}(2)$ if and only if it has no $U_{2,4}$ minor.*



Theorem 10.7 (Bixby 1979 / Seymour 1979). *A matroid is representable over $\text{GF}(3)$ if and only if it has no minor among $U_{2,5}$, $U_{3,5}$, F_7 , F_7^* .*



Theorem 10.8 (Tutte, 1958). *A matroid is representable over every field if and only if it has no minor among $U_{2,4}$, F_7 , F_7^* . Such matroids are called *regular*.*



Conjecture 10.9 (Rota, 1971). *For each finite field $\text{GF}(q)$ there is a finite list of excluded minors for $\text{GF}(q)$ -representability.*



Rota's conjecture is now a theorem: Geelen, Gerards, and Whittle announced a proof in 2014. The $\text{GF}(4)$ case (seven excluded minors) is due to Geelen, Gerards, and Kapoor (2000). Over $\mathbb{R}$, or any characteristic-zero field, there are infinitely many excluded minors (Lazarson, 1958). Which uniform matroids $U_{r,n}$ are representable over $\text{GF}(q)$ remains a lively question (the MDS conjecture).

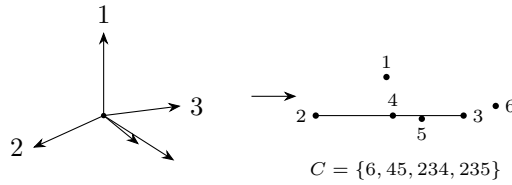
10.4 Non-linear matroids

F_7 is not representable in characteristic not 2; F_7^* is not representable in characteristic 2. Consequently:

Proposition 10.10. *The direct sum $F_7 \oplus F_7^*$ is not linear over any field: a representation would restrict to F_7 (forcing $\text{char} = 2$) and to F_7^* (forcing $\text{char} \neq 2$).*



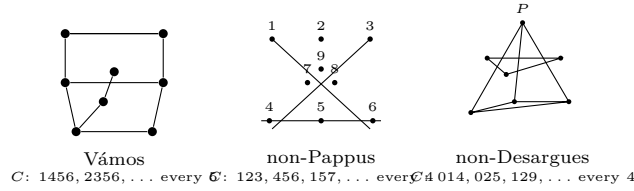
A configuration $v_1, \dots, v_n$ is *affinely independent* if $\sum a_i v_i = 0$ with $\sum a_i = 0$ forces all $a_i = 0$ (two points distinct, three non-collinear, four non-coplanar, ...).



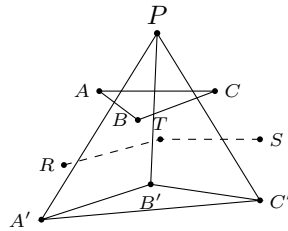
Proposition 10.11. *A loopless matroid is linear if and only if it is the affine matroid of some point configuration. (Pass to the linear matroid of the homogenised vectors $(1, v_i)$.)*



The non-Pappus, non-Desargues, and Vámos matroids arise by taking the indicated projective configurations and deleting a collinearity circuit; they are matroids, but not representable.



Theorem 10.12 (Desargues). *If triangles ABC and $A'B'C'$ are perspective from a point P , then the points $R = BC \cap B'C'$, $S = AC \cap A'C'$, $T = AB \cap A'B'$ are collinear.*



Proof in 3-space. Place the two triangles in different planes meeting in a line. Then AB and $A'B'$ meet on that line, and likewise for the other pairs, so R, S, T lie on the line of intersection of the two planes. $\square$

The *Desargues matroid* is the affine matroid of $\{A, B, C, A', B', C', P, R, S, T\}$; it is isomorphic to $M(K_5)$. The *non-Desargues matroid* is the same except that $\{R, S, T\}$ is declared independent. It is a matroid (direct verification of (I3), or the standard “relax a circuit-hyperplane” construction) but not representable, because any linear representation would force Desargues' theorem.

Two special cases of representability have names. A matroid is *binary* if it is representable over $\text{GF}(2)$, equivalently (Tutte) if it has no $U_{2,4}$ minor; graphic matroids are binary, which is why $U_{2,4}$ appears among Tutte's five forbidden minors for graphicness. A matroid is *regular* if it is representable

over every field, equivalently if it has a totally unimodular representation over $\mathbb{R}$ (every square submatrix has determinant in $\{0, \pm 1\}$). Regular matroids include all graphic and cographic matroids; Seymour's decomposition theorem (1980) builds every regular matroid from graphic ones, cographic ones, and a single 10-element sporadic R_{10} by 3-sums.

Whitney's 2-isomorphism theorem (the project “when is $M(G) \cong M(H)$?” on the board) says that two 2-connected graphs have isomorphic cycle matroids if and only if one is obtained from the other by a sequence of Whitney twists: cut along a 2-vertex separator and reverse one side.

Pappus's theorem is the companion of Desargues: given two lines with points A, B, C and A', B', C' respectively, the three points $AB' \cap A'B$, $AC' \cap A'C$, $BC' \cap B'C$ are collinear. Relaxing that collinearity circuit produces the non-Pappus matroid drawn above, which is not representable over any field (a coordinatisation would force Pappus). The same *circuit-hyperplane relaxation*—declaring independent a set that is simultaneously a circuit and a hyperplane, and taking the resulting new bases—is the standard way to manufacture non-representable matroids from representable ones, and is how non-Desargues arises from $M(K_5)$.

11 Algebraic matroids

Let $\mathbb{F} \subseteq \mathbb{K}$ be a field extension. An element $u \in \mathbb{K}$ is *algebraic* over $\mathbb{F}$ if $P(u) = 0$ for some $P \in \mathbb{F}[x]$, and *transcendental* otherwise. A finite $T \subseteq \mathbb{K}$ is algebraically dependent over $\mathbb{F}$ if some $t \in T$ is algebraic over $\mathbb{F}(T \setminus \{t\})$.

Theorem 11.1. *If $E \subseteq \mathbb{K}$ is finite, the algebraically independent subsets of E are the independent sets of a matroid.*



Definition 11.2. *A matroid is *algebraic* if it arises in this way.*



Example 11.3. Over $\mathbb{F} = \mathbb{R} \subseteq \mathbb{K} = \mathbb{C}(x, y, z)$, the elements

$$a = x, \quad b = z^2, \quad c = yz, \quad d = y, \quad e = 2y, \quad f = 1 + i$$

realise the running example. The explicit relations

$$f^2 - 2f + 2 = 0, \quad e - 2d = 0, \quad c^2 - d^2b = 0$$

show that f is a loop, $\{d, e\}$ is a parallel pair, and $\{b, c, d\}$, $\{b, c, e\}$ are 3-circuits, while a does not appear and is a coloop.



The board writes $b = (y + z)^2$ and $c = x(y + z)$ with $d = e = x$; that also makes $\{d, e\}$ parallel, but then a is entangled with b and is no longer a coloop. The specialisation above matches the circuits $\{f\}$, $\{d, e\}$, $\{b, c, d\}$, $\{b, c, e\}$ of the running example.

Example 11.4. Over $\mathbb{R} \subseteq \mathbb{C}$ every complex number is algebraic, so a finite subset of $\mathbb{C}$ consists entirely of loops. Over $\mathbb{Q} \subseteq \mathbb{R}$ the picture is mixed. For $\{a, b, c, d\} = \{\sqrt{2}, \sqrt{3}, e, e + \pi\}$ one has: a and b are loops ($t^2 - 2$, $t^2 - 3$); $c = e$ is transcendental (Hermite), hence independent; whether $d = e + \pi$ is algebraic over $\mathbb{Q}$, or over $\mathbb{Q}(e)$, is open.



To prove Theorem 11.1 we check that

$$\text{cl}(A) = \{x \in E : x \text{ is algebraic over } \mathbb{F}(A)\}$$

satisfies (CL1)–(CL4).

Lemma 11.5. *An element x is algebraic over $\mathbb{F}(a_1, \dots, a_n)$ if and only if the extension degree $[\mathbb{F}(a_1, \dots, a_n, x) : \mathbb{F}(a_1, \dots, a_n)]$ is finite.*



Proof. If $P(x) = 0$ has degree m , then $1, x, \dots, x^m$ are linearly dependent over $\mathbb{F}(a_1, \dots, a_n)$ and $1, x, \dots, x^{m-1}$ span $\mathbb{F}(a_1, \dots, a_n, x)$. If there is no algebraic relation then $1, x, x^2, \dots$ are linearly independent, so the degree is infinite. $\square$

Lemma 11.6 (Tower law). *If $\mathbb{F} \subseteq \mathbb{K} \subseteq \mathbb{L}$ then $[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{F}]$.*



Proof. If (x_i) is a $\mathbb{K}$ -basis of $\mathbb{L}$ and (y_j) is an $\mathbb{F}$ -basis of $\mathbb{K}$, then $(x_i y_j)$ is an $\mathbb{F}$ -basis of $\mathbb{L}$. $\square$

Proof of the closure axioms. (CL1): $a \in A$ is a root of $t - a \in \mathbb{F}(A)[t]$. (CL2): a polynomial over $\mathbb{F}(A)$ is a polynomial over $\mathbb{F}(B)$ if $A \subseteq B$. (CL4): if $a \in \text{cl}(A \cup \{b\})$ and $a \notin \text{cl}(A)$, a relation $P(a) = 0$ with coefficients in $\mathbb{F}(A, b)$ involves b (else $a \in \text{cl}(A)$). Write $P = \sum_{i=0}^m Q_i(b) a^i$ with $Q_i \in \mathbb{F}(A)[t]$ and some Q_i of positive degree; clearing the highest power of b produces a nonzero polynomial in b over $\mathbb{F}(A, a)$. Rearranging gives a polynomial equation for b over $\mathbb{F}(A, a)$, so $b \in \text{cl}(A \cup \{a\})$. (CL3): the inclusion $\text{cl}(A) \subseteq \text{cl}(\text{cl}(A))$ is (CL1). If $a \in \text{cl}(\text{cl}(A))$ then $[\mathbb{F}(\text{cl}(A), a) : \mathbb{F}(\text{cl}(A))] < \infty$. Each element of $\text{cl}(A)$ is algebraic over $\mathbb{F}(A)$, so adjoining them one at a time gives $[\mathbb{F}(\text{cl}(A)) : \mathbb{F}(A)] < \infty$. The tower law yields $[\mathbb{F}(\text{cl}(A), a) : \mathbb{F}(A)] < \infty$, hence $[\mathbb{F}(A, a) : \mathbb{F}(A)] < \infty$, so $a \in \text{cl}(A)$. $\square$

Thus algebraic independence is a matroid closure, and Theorem 11.1 follows from Theorem 7.7.

Every linear matroid is algebraic. Given a representation of M by vectors $v_1, \dots, v_n \in \mathbb{F}^r$, choose algebraically independent indeterminates $x_1, \dots, x_r$ over the prime field of $\mathbb{F}$, and set $e_i = \sum_j a_{ji} x_j$ using the coordinates of v_i . Algebraic independence of a subset of the e_i is then equivalent to linear independence of the corresponding v_i .

The converse is false. The non-Pappus matroid of the previous section is algebraic (Lindström) but not linear over any field. The Vámos matroid is neither linear nor algebraic: Ingleton's inequality, a linear rank inequality with no algebraic analogue of the same strength, already obstructs a representation by field elements. Thus

$$\text{graphic} \subsetneq \text{regular} \subsetneq \text{linear} \subsetneq \text{algebraic} \subsetneq \text{all matroids}.$$

Minors of algebraic matroids are algebraic, but it is open whether the dual of an algebraic matroid is always algebraic.

The rank of an algebraic matroid is the transcendence degree: if $E \subseteq \mathbb{K}$ is finite then $r(M) = \text{trdeg}(\mathbb{F}(E)/\mathbb{F})$. In the running realisation above one has $\mathbb{F}(E) = \mathbb{C}(x, y, z)$ of transcendence degree 3, matching $r(M) = 3$.